# JR Validated Environment — Process Capability Module

## User Guide for Process Capability Analysis

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JR Validated Environment

## Introduction

The Process Capability Module provides three validated R scripts for assessing whether a manufacturing process is capable of consistently producing parts or measurements within specification limits. Use these scripts to generate objective evidence for process validation packages (IQ/OQ/PQ) under 21 CFR 820.75 and ISO 13485 §7.5.6.

All three scripts are run through the JR validated environment (jrrun), which verifies project integrity and ensures the correct R library is used on every execution.

> **Validated environment:** Never run these scripts directly with Rscript. Always use the wrappers or jrrun so that the integrity check and controlled library are applied.

### Scripts in this module

| Script | Use when... | Output |
|---|---|---|
| **jrc_cap_normal** | Data is continuous and approximately normally distributed. | Histogram PNG |
| **jrc_cap_nonnormal** | Data is non-normal (skewed, bounded, or multi-modal). | Histogram PNG |
| **jrc_cap_sixpack** | You need a complete validation deliverable: stability + capability in one figure. | Six-panel PNG |

### Interpretation guide (applies to all scripts)

| Cpk / Ppk | Verdict | Interpretation |
|---|---|---|
| ≥ 1.67 | EXCELLENT | World-class. Typically requires no further action. |
| ≥ 1.33 | CAPABLE | Meets typical FDA / ISO 13485 expectation for process validation. |
| ≥ 1.00 | MARGINAL | Meets spec but with little margin. Consider process improvement. |
| < 1.00 | NOT CAPABLE | Process produces non-conformances. Action required before PQ. |

## Preparing Your Data

All three Cap scripts accept a two-column CSV file with a header row. The first column is typically a row identifier (id) and the second column contains the numeric measurement values.

### Required format

```
id,value
1,10.12
2,9.87
3,10.23
...
```

> **Column names:** The second column does not have to be named 'value'. Use the <col> argument to specify whichever column name your CSV uses (e.g. 'diameter', 'force', 'weight').

⚠ **Minimum sample size:** All Cap scripts require at least 5 valid observations. For reliable capability estimates, n ≥ 25 is recommended; n ≥ 100 is preferred for process validation.

### Specification limits

Both LSL (lower) and USL (upper) specification limits are optional individually, but at least one must be provided. Pass '-' to omit a limit:

- Two-sided: jrc_cap_normal data.csv value 9.0 11.0
- One-sided (USL only): jrc_cap_normal data.csv value - 11.0
- One-sided (LSL only): jrc_cap_normal data.csv value 9.0 -

## Script Reference

## 1. jrc_cap_normal — Capability Analysis (Normal)

### When do I use this?

Use this script when your measurement data is approximately normally distributed. It computes the standard capability indices (Cp, Cpk, Pp, Ppk, Cpm) using two estimates of process spread: within-subgroup sigma (estimated from the average moving range, suitable for individual measurements in time-order) and overall sample standard deviation. Both estimates are reported so you can compare short-term and long-term process performance.

### What do I need?

| Argument | What to provide |
|---|---|
| **data.csv** | Path to the two-column CSV file with your measurements. |
| **<col>** | Name of the numeric column to analyse (e.g. value, diameter). |
| **<lsl>** | Lower Specification Limit (numeric), or '-' to omit. |
| **<usl>** | Upper Specification Limit (numeric), or '-' to omit. |

### Example

```
jrc_cap_normal measurements.csv diameter 9.90 10.10
```

### What to look for in the output

Check the terminal for the Cpk and Ppk values and the capability verdict. A histogram PNG is saved to ~/Downloads/ showing your data distribution (bars), the fitted normal curve (blue line), spec limits (red dashed lines), and the process mean (dark vertical line). For process validation: Cpk ≥ 1.33 confirms capability at the typical FDA / ISO 13485 threshold.

### Output index reference

| Index | Sigma used | Interpretation |
|---|---|---|
| Cp | Within (σ_w = MR-bar/d2) | Potential — how good could the process be if perfectly centred? |
| Cpk | Within (σ_w = MR-bar/d2) | Actual short-term capability, accounting for mean position. |
| Cpm | Overall (s) | Taguchi index — penalises deviation from target (spec |

| | | |
|---|---|---|
| | | midpoint). |
| Pp | Overall (s) | Potential — long-term spread relative to spec width. |
| Ppk | Overall (s) | Actual long-term performance, accounting for mean position. |

> ℹ **Cp vs Cpk:** If Cp >> Cpk the spread is adequate but the mean is off-target. Address process centring before concluding on capability.

> ℹ **Cpk vs Ppk:** If Ppk << Cpk there is long-term drift or batch-to-batch variation beyond the short-term noise. Investigate assignable causes.

## 2. jrc_cap_nonnormal — Capability Analysis (Non-Normal)

### When do I use this?

Use this script when your data is not normally distributed — for example, torque values that are bounded below zero, particle counts, or any measurement that shows clear skew or multi-modality. It uses the percentile method (ISO 22514-2 / AIAG SPC Manual): the 0.135th and 99.865th sample percentiles are used as the ±3σ equivalents, making no distributional assumption. The script also runs a Shapiro-Wilk normality test and will advise you to use jrc_cap_normal instead if the data appears approximately normal.

### What do I need?

| Argument | What to provide |
|---|---|
| **data.csv** | Path to the two-column CSV file with your measurements. |
| **<col>** | Name of the numeric column to analyse. |
| **<lsl>** | Lower Specification Limit (numeric), or '-' to omit. |
| **<usl>** | Upper Specification Limit (numeric), or '-' to omit. |

### Example

```
jrc_cap_nonnormal torque_data.csv torque 4.5 5.5
```

### What to look for in the output

The terminal shows Pp and Ppk (percentile method), observed non-conformance percentages, and the Shapiro-Wilk p-value. The histogram PNG shows the data distribution with a KDE overlay (orange curve), spec limits (red dashed), and P0.135/P99.865 markers (blue dotted). Ppk ≥ 1.33 is the standard acceptance criterion for non-normal process validation.

> ⚠ **Sample size for percentile method:** The percentile method estimates extreme quantiles (0.135th, 99.865th) from the sample. At n < 30 these estimates are unreliable. For production process validation, collect n ≥ 100 before relying on the percentile Ppk.

## 3. jrc_cap_sixpack — Capability Sixpack

### When do I use this?

Use this script to produce the standard process capability deliverable for a process validation package. The Sixpack combines process stability evidence (I-MR control charts) with capability evidence (histogram, Q-Q plot, indices) in a single 3600×2400 px figure suitable for direct inclusion in a validation report. Run it after confirming approximate normality — it applies the same within-sigma Cpk / overall-sigma Ppk methodology as jrc_cap_normal.

### What do I need?

| Argument | What to provide |
|---|---|
| **data.csv** | Path to the two-column CSV file with your measurements. |
| **<col>** | Name of the numeric column to analyse. |
| **<lsl>** | Lower Specification Limit (numeric), or '-' to omit. |
| **<usl>** | Upper Specification Limit (numeric), or '-' to omit. |

### Example

```
jrc_cap_sixpack validation_data.csv value 9.0 11.0
```

### What to look for in the output

The terminal echoes all computed values. The PNG (saved to ~/Downloads/) shows six panels: top row — Individuals chart and MR chart; middle row — histogram and normal probability plot; bottom row — numerical summary table and colour-coded verdict panel (green = capable, amber = marginal, red = not capable). A capable, in-control process shows no OOC signals, histogram well inside spec limits, Q-Q points close to the reference line, and Cpk ≥ 1.33.

### Panel descriptions

| Panel | What it shows |
|---|---|
| Individuals (X) chart | Each measurement plotted in time-order with UCL/LCL (±3σ_w). Spec limits overlaid as dotted lines. OOC points highlighted in red. |
| Moving Range chart | Absolute differences between consecutive observations, with UCL = 3.267×MR-bar. Detects sudden step changes or inflating variability. |
| Histogram | Frequency distribution of all measurements with normal curve overlay (based on overall sigma) and spec limit lines. |
| Normal probability plot | Q-Q plot against a normal reference line. Points close to the line support the normality assumption. Shapiro-Wilk p-value in title. |
| Summary table | All computed indices: n, X-bar, σ_w, s, Cp, Cpk, Cpm, Pp, Ppk, sigma level, and estimated PPM out-of-specification. |
| Verdict panel | Capability verdict with colour background: green (Cpk ≥ 1.33), amber (1.00 ≤ Cpk < 1.33), red (Cpk < 1.00). |

## Typical Process Validation Workflow

### Step 1 — Check your data format

Prepare a two-column CSV (id, value). Verify there are no text values or blanks in the measurement column. n ≥ 25 recommended.

### Step 2 — Check normality

Run jrc_normality (community module) or jrc_cap_nonnormal to check the Shapiro-Wilk result. If p ≥ 0.05 and the histogram looks symmetric, proceed to jrc_cap_normal or jrc_cap_sixpack.

### Step 3 — Run capability analysis

```
jrc_cap_sixpack mydata.csv value 9.0 11.0
```

Check terminal for the Cpk value and verdict. Open the PNG in ~/Downloads/.

### Step 4 — Interpret and record

If Cpk ≥ 1.33 and the I-MR chart shows no OOC signals, the process is both stable and capable. Include the Sixpack PNG in your validation report as objective evidence under 21 CFR 820.75 / ISO 13485 §7.5.6.

⚠️ **OOC signals on the I-MR chart:** If the Individuals or MR chart signals OOC points, the process is not in statistical control. Investigate assignable causes before interpreting capability indices — a Cpk computed from an out-of-control process is not a reliable predictor of future performance.

✅ **One-sided analyses:** For characteristics with only one meaningful spec limit (e.g. minimum peel strength, maximum contamination), pass '-' for the unused limit. Cp/Pp and Cpm require both limits and will not be reported.

## Troubleshooting

| Error / symptom | Resolution |
|---|---|
| RENV_PATHS_ROOT is not set | Run via jrrun or the wrapper, not Rscript directly. Example: jrrun jrc_cap_normal.R data.csv value 9.0 11.0 |
| File not found | Check the path to your CSV. Use an absolute path if the file is not in the current directory. |
| Column not found in CSV | Check column names in the CSV header. The column name is case-sensitive. |
| Need at least 5 observations | Increase your sample size. Check for blank or non-numeric rows in the CSV. |
| LSL must be less than USL | Check that you have not swapped the spec limit arguments. |
| Cpk is very low but data looks fine | If within-sigma (σ_w) >> overall sigma (s), the MR chart may have OOC signals inflating the moving-range estimate. Check the Sixpack I-MR chart. |
| Shapiro-Wilk advisory appears | jrc_cap_nonnormal detected SW p ≥ 0.05 (data appears normal). Consider switching to jrc_cap_normal for a Cpk analysis. |

## References


AIAG SPC Manual, 2nd Edition (2005). Automotive Industry Action Group.

ISO 22514-2:2017. Statistical methods in process management — Capability and performance. Part 2: Process capability and performance of time-dependent process models.

Kane VE (1986). Process capability indices. Journal of Quality Technology, 18(1), 41–52.

Taguchi G, Elsayed EA, Hsiang TC (1989). Quality Engineering in Production Systems. McGraw-Hill.

21 CFR Part 820, §820.75. Quality System Regulation — Process Validation.

ISO 13485:2016, §7.5.6. Medical devices — Quality management systems.